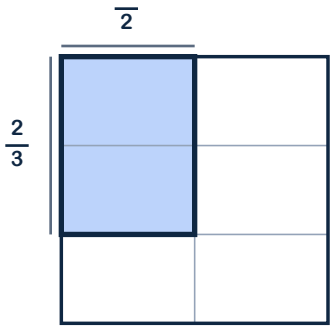
Multiplying Fractions

A fraction of a fraction: cut one way, then the other way.

1/2 × 2/3: half of two thirds



Cut the square into 3 rows: shade 2 of them (2/3).
Cut it into 2 columns: take half of the shaded part.
The square now has 6 pieces. 2 are in both.

$$\frac{1}{2} \times \frac{2}{3} = \frac{2}{6}$$

Without a picture: tops times tops, bottoms times bottoms

$$\frac{2}{3} \times \frac{4}{5} = \frac{8}{15}$$

2 × 4 = 8 pieces are in both.
3 × 5 = 15 pieces in the whole square.

Smaller or bigger?

$$\frac{3}{4} \times 8 = 6$$

Times a fraction less than 1: the answer is smaller than 8.

$$1\frac{1}{2} \times 8 = 12$$

Times more than 1: the answer is bigger than 8.

Your turn!

$$\frac{1}{3} \times \frac{3}{4} = \frac{\square}{\square}$$

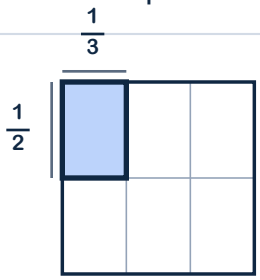
Multiplying Fractions: Practice

Read × as "of". Tops times tops, bottoms times bottoms.

Part 1 Read the picture

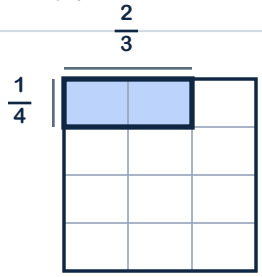
How many pieces are in both? How many pieces in all?

1)



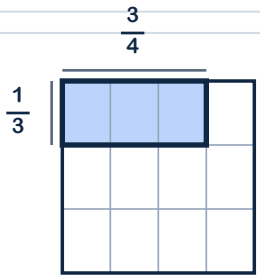
$\frac{1}{3} \times \frac{1}{2} = \frac{\square}{\square}$

2)



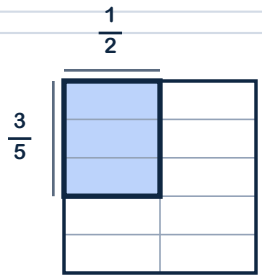
$\frac{2}{3} \times \frac{1}{4} = \frac{\square}{\square}$

3)



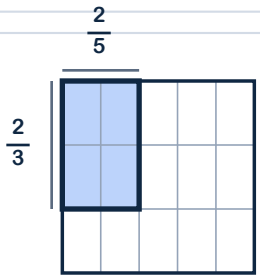
$\frac{3}{4} \times \frac{1}{3} = \frac{\square}{\square}$

4)



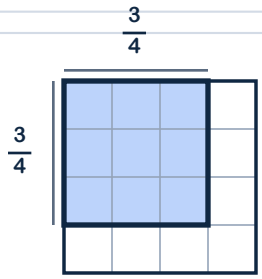
$\frac{1}{2} \times \frac{3}{5} = \frac{\square}{\square}$

5)



$\frac{2}{5} \times \frac{2}{3} = \frac{\square}{\square}$

6)

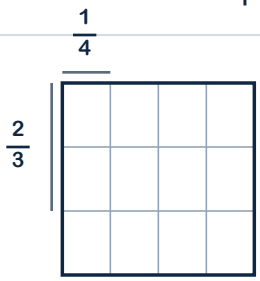


$\frac{3}{4} \times \frac{3}{4} = \frac{\square}{\square}$

Part 2 Shade and multiply

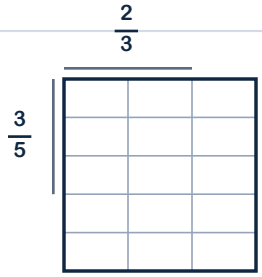
Shade the part that is in both, then write it.

7)



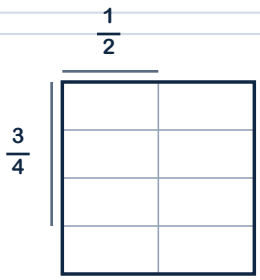
$\frac{1}{4} \times \frac{2}{3} = \frac{\square}{\square}$

8)



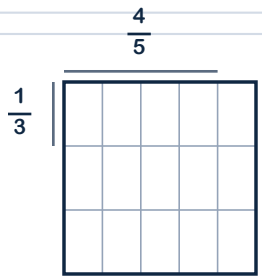
$\frac{2}{3} \times \frac{3}{5} = \frac{\square}{\square}$

9)



$\frac{1}{2} \times \frac{3}{4} = \frac{\square}{\square}$

10)



$\frac{4}{5} \times \frac{1}{3} = \frac{\square}{\square}$

Multiplying Fractions: Practice (continued)

Part 3 Multiply No help this time.

11) $\frac{2}{3} \times \frac{4}{5} = \frac{\square}{\square}$

12) $\frac{1}{4} \times \frac{3}{7} = \frac{\square}{\square}$

13) $\frac{3}{5} \times \frac{2}{9} = \frac{\square}{\square}$

14) $\frac{5}{6} \times \frac{1}{3} = \frac{\square}{\square}$

15) $\frac{3}{8} \times \frac{1}{2} = \frac{\square}{\square}$

16) $\frac{2}{7} \times \frac{3}{4} = \frac{\square}{\square}$

17) $\frac{4}{9} \times \frac{1}{5} = \frac{\square}{\square}$

18) $\frac{1}{6} \times \frac{5}{8} = \frac{\square}{\square}$

19) $\frac{3}{10} \times \frac{1}{4} = \frac{\square}{\square}$

20) $\frac{5}{7} \times \frac{2}{3} = \frac{\square}{\square}$

21) $\frac{1}{5} \times \frac{4}{5} = \frac{\square}{\square}$

22) $\frac{7}{8} \times \frac{1}{6} = \frac{\square}{\square}$

Multiplying Fractions: Practice (continued)

Part 3 Multiply (continued)

23) $\frac{2}{9} \times \frac{2}{5} = \frac{\square}{\square}$

24) $\frac{3}{4} \times \frac{5}{6} = \frac{\square}{\square}$

25) $\frac{1}{8} \times \frac{3}{10} = \frac{\square}{\square}$

26) $\frac{6}{7} \times \frac{1}{2} = \frac{\square}{\square}$

Part 4 A fraction of a number No help this time.

27) $\frac{2}{3} \times 9 = \square$

28) $\frac{3}{4} \times 12 = \square$

29) $\frac{4}{5} \times 15 = \square$

30) $\frac{5}{6} \times 18 = \square$

31) $\frac{3}{8} \times 56 = \square$

32) $\frac{4}{7} \times 14 = \square$

Multiplying Fractions: Practice (continued)

Part 4 A fraction of a number (continued)

33) $\frac{1}{6} \times 30 = \square$

34) $\frac{7}{10} \times 20 = \square$

35) $\frac{3}{5} \times 30 = \square$

36) $\frac{5}{8} \times 24 = \square$

Part 5 Mixed numbers times a number No help this time. Wholes first, then the piece.

37) $1\frac{1}{2} \times 4 = \square$

38) $2\frac{1}{3} \times 6 = \square$

39) $1\frac{3}{4} \times 4 = \square$

40) $3\frac{1}{2} \times 6 = \square$

41) $2\frac{2}{5} \times 5 = \square$

42) $2\frac{3}{4} \times 4 = \square$

43) $1\frac{1}{6} \times 6 = \square$

44) $4\frac{1}{3} \times 9 = \square$

Multiplying Fractions: Practice (continued)

Part 6 Smaller or bigger? No help this time. Write < or >. Don't work it out: think.

45) $\frac{3}{4} \times 8 \square 8$

46) $1\frac{1}{2} \times 10 \square 10$

47) $\frac{2}{3} \times 9 \square 9$

48) $2\frac{1}{4} \times 20 \square 20$

49) $\frac{5}{6} \times 12 \square 12$

50) $1\frac{3}{5} \times 7 \square 7$

51) $\frac{7}{8} \times 16 \square 16$

52) $1\frac{1}{3} \times 15 \square 15$

53) $\frac{4}{9} \times 18 \square 18$

54) $3\frac{2}{7} \times 6 \square 6$

Multiplying Fractions: Practice (continued)

Part 7 Multiply

55) $\frac{2}{3} \times \frac{2}{5} = \frac{\square}{\square}$

56) $\frac{3}{6} \times \frac{1}{3} = \frac{\square}{\square}$

57) $\frac{1}{4} \times \frac{8}{12} = \frac{\square}{\square}$

58) $\frac{7}{10} \times \frac{1}{2} = \frac{\square}{\square}$

59) $\frac{1}{8} \times \frac{8}{10} = \frac{\square}{\square}$

60) $\frac{8}{10} \times \frac{4}{6} = \frac{\square}{\square}$

61) $\frac{4}{10} \times \frac{7}{10} = \frac{\square}{\square}$

62) $\frac{5}{8} \times \frac{2}{5} = \frac{\square}{\square}$

63) $\frac{6}{10} \times \frac{1}{12} = \frac{\square}{\square}$

64) $\frac{2}{3} \times \frac{9}{12} = \frac{\square}{\square}$

65) $\frac{1}{2} \times \frac{3}{4} = \frac{\square}{\square}$

66) $\frac{11}{12} \times \frac{4}{6} = \frac{\square}{\square}$

Multiplying Fractions: Practice (continued)

Part 7 Multiply (continued)

67) $\frac{1}{10} \times \frac{2}{4} = \frac{\square}{\square}$

68) $\frac{2}{4} \times \frac{3}{4} = \frac{\square}{\square}$

69) $\frac{5}{12} \times \frac{5}{8} = \frac{\square}{\square}$

70) $\frac{2}{12} \times \frac{2}{3} = \frac{\square}{\square}$

71) $\frac{1}{4} \times \frac{1}{12} = \frac{\square}{\square}$

72) $\frac{3}{12} \times \frac{1}{2} = \frac{\square}{\square}$

73) $\frac{1}{2} \times \frac{5}{6} = \frac{\square}{\square}$

74) $\frac{4}{6} \times \frac{1}{3} = \frac{\square}{\square}$

75) $\frac{11}{12} \times \frac{4}{5} = \frac{\square}{\square}$

76) $\frac{1}{2} \times \frac{5}{10} = \frac{\square}{\square}$

77) $\frac{1}{2} \times \frac{7}{8} = \frac{\square}{\square}$

78) $\frac{1}{10} \times \frac{4}{6} = \frac{\square}{\square}$

Multiplying Fractions: Practice (continued)

Part 8 A fraction of a number

79) $\frac{2}{6} \times 18 = \square$

80) $\frac{1}{2} \times 12 = \square$

81) $\frac{2}{4} \times 8 = \square$

82) $\frac{1}{3} \times 6 = \square$

83) $\frac{2}{3} \times 6 = \square$

84) $\frac{1}{3} \times 15 = \square$

85) $\frac{3}{6} \times 24 = \square$

86) $\frac{1}{2} \times 8 = \square$

87) $\frac{4}{5} \times 20 = \square$

88) $\frac{1}{2} \times 10 = \square$

89) $\frac{2}{3} \times 15 = \square$

90) $\frac{1}{5} \times 10 = \square$

Multiplying Fractions: Practice (continued)

Part 8 A fraction of a number (continued)

91) $\frac{2}{6} \times 24 = \square$

92) $\frac{1}{4} \times 8 = \square$

93) $\frac{7}{8} \times 40 = \square$

94) $\frac{1}{3} \times 12 = \square$

95) $\frac{2}{3} \times 18 = \square$

96) $\frac{1}{6} \times 18 = \square$

Part 9 Remember? Problems from earlier packets.

97) $\frac{1}{3} + \frac{1}{2} = \frac{\square}{\square}$

98) $\frac{4}{5} - \frac{1}{4} = \frac{\square}{\square}$

99) $6 \times 0.7 = \square$

100) $0.4 \times 0.6 = \square$

101) $3 \times \frac{2}{7} = \frac{\square}{7}$

102) $752 \div 16 = \square$

Multiplying Fractions: Practice (continued)

Part 10 Word problems × means "of".

103) A recipe needs 3/4 cup of milk. Mia makes half of the recipe. How much milk does she need?

Number sentence _____

Answer ☐ cup

Draw or show your work

104) A garden is 2/3 of the yard. Tomatoes grow on 1/2 of the garden. What part of the yard is that?

Number sentence _____

Answer ☐ of the yard

Draw or show your work

105) A class has 24 kids. 3/8 of them walk to school. How many kids walk?

Number sentence _____

Answer ☐ kids

Draw or show your work

106) A ribbon is 1 1/2 m long. Leo needs 6 ribbons. How much ribbon is that?

Number sentence _____

Answer ☐ m

Draw or show your work

107) Challenge: Ava had 4/5 of a pizza. She ate 1/4 of that. How much of the whole pizza did she eat?

Number sentence _____

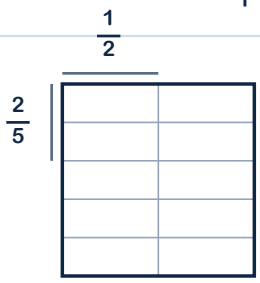
Answer ☐ of the pizza

Draw or show your work

Check-In: Multiplying Fractions

Show what you know. No help this time. Score: _____ / 11

Part 1 Shade and multiply

1) 

$\frac{1}{2} \times \frac{2}{5} = \frac{\square}{\square}$

Part 2 Multiply

2) $\frac{2}{5} \times \frac{3}{7} = \frac{\square}{\square}$

3) $\frac{3}{4} \times \frac{2}{5} = \frac{\square}{\square}$

4) $\frac{1}{2} \times \frac{7}{9} = \frac{\square}{\square}$

5) $\frac{4}{7} \times \frac{2}{3} = \frac{\square}{\square}$

Part 3 A fraction of a number

6) $\frac{3}{4} \times 20 = \square$

7) $\frac{5}{6} \times 24 = \square$

8) $\frac{3}{5} \times 10 = \square$

Part 4 Smaller or bigger? Write < or >.

9) $\frac{2}{5} \times 10 \ \square \ 10$

10) $1\frac{1}{4} \times 8 \ \square \ 8$

11) $\frac{5}{7} \times 14 \ \square \ 14$